



The Spaghetti Engine
VISUAL SYSTEM MODELLING WORKSPACE

The Spaghetti Engine

Development Log

Fuzzy Cognitive Map modelling tool

Field	Value
Version	0.4.3 (prototype)
Log entry date	10 October 2026
Status	Released: pushed to the GitHub main branch in the commit that adds this log; Netlify deploys the site automatically.
Live URL	spaghetti-engine.netlify.app (Netlify, auto-deployed from the GitHub main branch)
Source repository	github.com/aPaarl/Spaghetti-engine, main branch (commit titled Release v0.4.3)
Previous version	0.4.2, 9 October 2026, commits d98d0ef and b33738c (see Spaghetti_Engine_Dev_Log_v0.4.2.docx)
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1. Purpose of This Log

This Development Log records the changes from version 0.4.2 (9 October 2026) to version 0.4.3 of The Spaghetti Engine. Version 0.4.3 changes how transition points are defined on the Transition Point and Transition Pathways tabs: a transition point is now the intensity from which the marginal benefit declines most steeply, the point of diminishing returns, instead of the steepest step of the response. The runs behind both tabs are settled far more tightly, because the new definition compares neighbouring changes of a curve. The definition was chosen for the analysis of the Brazilian Amazon case (Paper 2), which reports transition points computed this way.

The log documents what changed and why, how it was checked against independently computed values, how existing results are affected, and the deployment status.

2. Summary of Version 0.4.3

Area	Change
Transition point	The intensity from which the marginal benefit declines most steeply: $MB(i) = s(Y(i+1) - Y(i)) / \Delta x$, and the transition point is $x(i)$ at the minimum of $(MB(i) - MB(i-1)) / \Delta x$ over the interior levels. It is 1 when that minimum is at the last interior level, or when the marginal benefit does not decline at all.
Response types	Flat, Early (TP at most 1/3), Middle, Late (TP of 2/3 or more), and Proportional (TP = 1) replace Front-Loaded, Threshold-Type, and Back-Loaded.
Results table	Change at TP is now $Y(TP) - Y(0)$, and a new column, Share at TP, gives it as a share of Max Change. The marginal-return chart becomes a marginal-benefit chart with the transition points marked.
Convergence	Every run of both transition tabs continues until no concept changes by more than $1e-9$ per iteration, with at least 1000 iterations. Whether a run counts as converged still follows the global threshold; both counts are shown.
Transition Pathways	A stage stops at the marginal-benefit transition point of the total outcome gain, and a lever is only added if the gain there is higher than the gain already reached. The greedy search ranks by the largest marginal benefit, as before.
Record of 0.4.2	Version 0.4.2 was pushed as commits d98d0ef and b33738c on 9 October 2026 and deployed the same day.

3. The Changes in Detail

3.1 The marginal-benefit transition point

- **Definition.** For a response Y sampled at intensities $x(i) = i/n$, the marginal benefit of step i is $MB(i) = s(Y(i+1) - Y(i)) / \Delta x$, with s the direction that counts as benefit. The transition point is the sampled level $x(i)$, $i = 1$ to $n-1$, at

which $(MB(i) - MB(i-1)) / \Delta x$ is lowest: the point from which each further step of effort adds markedly less than the step before.

- **Why.** The steepest step marks where an outcome changes fastest, not where extra effort stops paying off. For a saturating response the steepest step lies at the very start, so the old definition put the transition point of almost every concave response at 0; the new one marks where the returns start to diminish, which is the question a decision maker asks about the intensity of an intervention.
- **No decline.** When the steepest decline falls on the last interior level, or the marginal benefit does not decline at all (a straight or ever-steepening response), the benefit keeps accruing up to full intensity and the transition point is 1: the Proportional response type. A decline smaller than 1% of the largest marginal benefit per unit of intensity counts as none, so rounding noise on a straight response does not create a transition point.
- **Orientation.** The Transition Point tab has no notion of a desirable direction, so s is the sign of the outcome's total effect over the sweep. The Transition Pathways tab uses the total outcome gain, which already counts every outcome in its desirable direction, so $s = +1$ there.

3.2 Transition Point tab

- **Results.** The table gains Share at TP; Change at TP is the change up to the transition point; the response types are Flat, Early, Middle, Late, and Proportional. Share at TP shows n/a when Max Change is below the flat-response threshold, where a share says nothing.
- **Charts.** The second chart shows each outcome's marginal benefit, in the direction the outcome moves, with dashed lines at the transition points; its CSV and PNG exports are named transition-marginal-benefit. The response curves keep their dashed transition point lines.
- **Dashboard.** The earliest and latest transition point cards leave flat outcomes out, unless every outcome is flat, because a flat outcome's transition point means nothing. Efficiency is unchanged: total effect divided by the transition point.
- **Text.** The explanation box, Method & diagnostics, the help texts, and the Analysis Workbook sheet follow the new definition; the workbook sheet and the CSV export have a Share at TP column.

3.3 Convergence of the transition runs

- **Run on to 1e-9.** The transition point is a second difference of the response curve, which multiplies whatever is left of a run that has not quite settled by the square of the resolution. Every run of both transition tabs therefore continues until no concept changes by more than $1e-9$ per iteration (or the global threshold, if smaller), with at least 1000 iterations (or the global limit, if higher).
- **Converged.** Whether a run counts as converged still follows the global threshold, so a model that settled before still counts as settled. Method & diagnostics shows both counts: runs converged, and runs settled to $1e-9$.
- **Cost.** Models that settle take a fraction of a second per sweep (the peat meadow example under the sigmoid and absolute rule: 25 ms at 500 increments). A model that keeps oscillating runs to the limit at every step: the water-energy-food example under tanh and the relative rule took 3.2 seconds at 500 increments, with 381 of 501 runs not converged.

3.4 Transition Pathways tab

- **Stages.** Each stage stops at the marginal-benefit transition point of the total outcome gain along the lever's sweep from the state reached so far. A lever is added only if the gain at that point exceeds the gain the pathway has already reached, so a stage under the default rule always raises the total gain.

- **Lever profiles.** Each lever's stand-alone transition point and each outcome's own transition point follow the same definition; the response types are those of the Transition Point tab, or No improvement.
- **Iterations.** Pathway runs use at least 1000 iterations (the default of the setting is now 1000, and a remembered setup with the old 500 is raised to 1000). Method & diagnostics reports how many runs settled to 1e-9.

4. Implementation Choices

Topic	Choice in the tool	Reason
Where the transition point lies	At $x(i)$, the level between the two steps whose marginal benefits differ most.	This is the definition used in Paper 2. Its values were reproduced exactly (Section 7).
Last interior level and no decline	Reported as 1 (Proportional).	A decline only at the last step, or none at all, means the benefit keeps accruing up to full intensity; reporting the last interior level would suggest a point of diminishing returns that is not there. Without this rule a convex response would get the transition point of its smallest increase, which can be at the start.
Noise tolerance	A decline below 1% of the largest marginal benefit per unit of intensity counts as none.	Second differences of a straight response are rounding noise; at 500 increments that noise is of the order of 0.1 to 1% of the marginal benefit when runs settle to 1e-9.
Convergence	Run on to 1e-9; count as converged by the global threshold.	The precision the definition needs, without turning every model that settled at the old threshold into a non-converged one. Both counts are reported.
When a lever takes part	The gain at the transition point must exceed the gain already reached.	Comparing with the lever held at 0 would admit a stage that leaves the pathway worse off when holding the lever at 0 itself moves the system.
Share at TP	n/a when Max Change is below the flat-response threshold.	A share of a negligible total effect produces meaningless numbers, such as tens of thousands of percent.
Efficiency	Unchanged: total effect divided by the transition point.	Paper 2 uses it with the new transition point.

5. Effect on Existing Results

- Transition points from 0.4.2 and earlier mark the steepest step and are not comparable with those of this version; re-run the analysis. For S-shaped responses they are often close, for saturating responses very different (from 0 to the point where the returns fall).
- Transition Pathways results change: stages stop at different intensities, and some levers are added or left out differently. In the peat meadow example (Paper 3) with food production, emissions, and biodiversity as

outcomes and three stages, the five levers keep their stand-alone transition points at 20 steps, but the analysis finds 61 instead of 71 pathways, 8 instead of 10 admissible ones, and no Deep Investment pathway. Results published with 0.4.2 and earlier should be recomputed with this version before they are compared.

- The response type names change, the Transition Point CSV export and workbook sheet have a Share at TP column after Change at TP, and the marginal-benefit CSV and PNG are named transition-marginal-benefit instead of transition-marginal-returns.
- Version raised from 0.4.2 to 0.4.3 in package.json and package-lock.json.

6. Defects Found and Fixed

Problem	Cause	Fix
The Transition Point tab reported transition points from runs that had not settled, without saying so: on the water-energy-food example under the default settings, none of the 101 runs of a sweep of Renewable energy capacity converged within the 100 iterations.	The sweep did not count non-converged runs.	The tab now counts runs that did not converge (and runs that did not reach 1e-9) and says what that means.
During development, Share at TP showed values such as 46505% for outcomes that end where they started.	A share of a total effect of nearly zero.	n/a below the flat-response threshold.

7. Verification

- **Reference values:** the Paper 2 transition points, computed independently from the same model (12 levers and 3 outcomes, sigmoid with λ 1, absolute rule, at 500 and 100 increments), were reproduced exactly: 72 of 72, including the shares at the transition point.
- **Tests:** tests/test_marginal_benefit_tp.cjs (new, 37 checks): the definition on curves with a known answer (an S-curve against its analytic point of steepest decline, a straight line with and without rounding noise, a convex and a saturating response, a decline at the last level, a flat response), the classification, the sweep settings, and the pathway rules. tests/test_transition_direction.cjs (37 checks) now checks Change at TP as $Y(TP) - Y(0)$ and Share at TP. All pass.
- **Browser:** the Transition Point and Transition Pathways tabs run and show the new columns, response types, counts, and charts; no new console errors.
- **Build:** the production build completes without errors.

8. Deployment Status

- **Deployed** with the push of this release to main (commit titled Release v0.4.3). Netlify rebuilds the site automatically.
- **Files changed:** src/SpaghettiEngine.jsx, tests/test_transition_direction.cjs, tests/test_marginal_benefit_tp.cjs (new), tools/docx2pdf.py, package.json and package-lock.json, README.md, src/docs-manifest.json, public/sitemap.xml, and the three PDFs in public/docs, where the 0.4.3 versions replace those of 0.4.2.
- **Build note:** npm run build fails in the local OneDrive folder because npm's command shim cannot handle the ampersand in the folder path; running node node_modules/vite/bin/vite.js build directly works.
- **PDF note:** Edge's headless printing produced no file on the build machine for this release, so the PDFs were printed with Chrome (EDGE_PATH set to chrome.exe); tools/docx2pdf.py now also gives the browser a profile folder of its own, so that an open browser window cannot take the call over.

9. Prototype Scope and Known Limitations

- All limitations listed in the 0.4.0 to 0.4.2 logs still apply, except that Change at TP no longer includes a step.
- The transition point is local. When the marginal benefit dips early and rises again later (a concave, then convex response), it marks the early dip, even if the returns later increase. In the peat meadow example, raised groundwater levels at 0.45 followed by area-based governance gives area-based governance a transition point of 0.05, although its marginal benefit later more than doubles.
- A model that keeps oscillating cannot settle to 1e-9, and its transition points stay imprecise; the tab says so.
- The transition point depends on the sweep resolution: re-run at a higher resolution before quoting a value.
- The regression tests of 0.4.0 are still not in the repository; tests/ holds the two test files named above.

10. Related Documentation

Document	Covers
The Spaghetti Engine - User Manual (version 0.4.3)	How to use every feature. Section 1.2 lists the changes of 0.4.3; Sections 12 and 13 describe the transition point, and Section 12.6 the precision of the runs.
The Spaghetti Engine - Methodological Annex (version 0.4.3)	The methods. Section 12.1 defines the marginal-benefit transition point, Section 13.3 the conditional transition point, and Section 13.10 the convergence of the runs.
Spaghetti_Engine_Dev_Log_v0.4.2	The 0.4.2 release: readable text and the Change at TP column.
Spaghetti_Engine_Dev_Log_v0.4.1	The 0.4.1 release: a decrease runs from 0 to -1 on the Transition Point and Transition Pathways tabs.
This Development Log	What changed in 0.4.3 and the technical work behind it.

11. Next Steps

- Decide whether a stage should look past an early dip in the marginal benefit when it rises again later (Section 9).
- Rebuild the full engine regression set inside tests/.
- Continue with the development roadmap: version 0.5.0 (guided first use) is next.